

Stay or Switch? Strategic Memory Portability in Competing AI Assistants

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REVIEW-READY V1

KEY MESSAGE Portability can be socially valuable yet strategically absent. At the benchmark, lock-in and low-type portability are both pure Bayesian equilibria, while user switching remains an untested behavioral question.

01 Question & model

Research question & verified gap

Shared question: When do competing AI assistants with private implementation costs supply portable memory, and when does that availability improve user mobility?

Verified gap: Switching-cost and compatibility research explains lock-in, but does not jointly test this Bayesian game, its welfare and incentive comparisons, and AI-user choice [1, 3].

Strategic model & foundations

Players and timing: Two platforms privately observe cost c_i in $\{.2, 1.2\}$, with $\Pr(c_i=.2)=.7$, then simultaneously choose Portable (P) or Locked (L).

Payoffs: $EU(P|c)=3q-1-c$ and $EU(L|c)=q$.

Solution concept: Bayesian Nash equilibrium fits private types [2]. Direct enumeration checks all 16 strategy profiles.

Three complementary lenses

Game theory: Exactly two pure BNE, (LL,LL) and (PL,PL). The low-type mixed probability is $6/7$.

Social choice: At $m=1$, welfare is 0, 2.17, and 4 for LL, PL, and PP. PP is a comparator, not a BNE.

Mechanism design: At $p=.4$, benefit $\tau=.4$ weakly sustains PL. LL remains weak through $\tau=1.2$.

Auction or allocation comparison

Mapping: One primary-assistant slot and two iid $U[0,1]$ platform values.

Rules: First price uses $b(v;r)=(v^2+r^2)/(2v)$. Second price uses truthful bids [4].

Comparison: Lowering the assumed reserve from .50 to .20 raises allocation from .75193 to .96004. Conditional efficiency is 1.

Boundary: This is an allocation analogy. The user is not auctioned.

02 Design & evidence

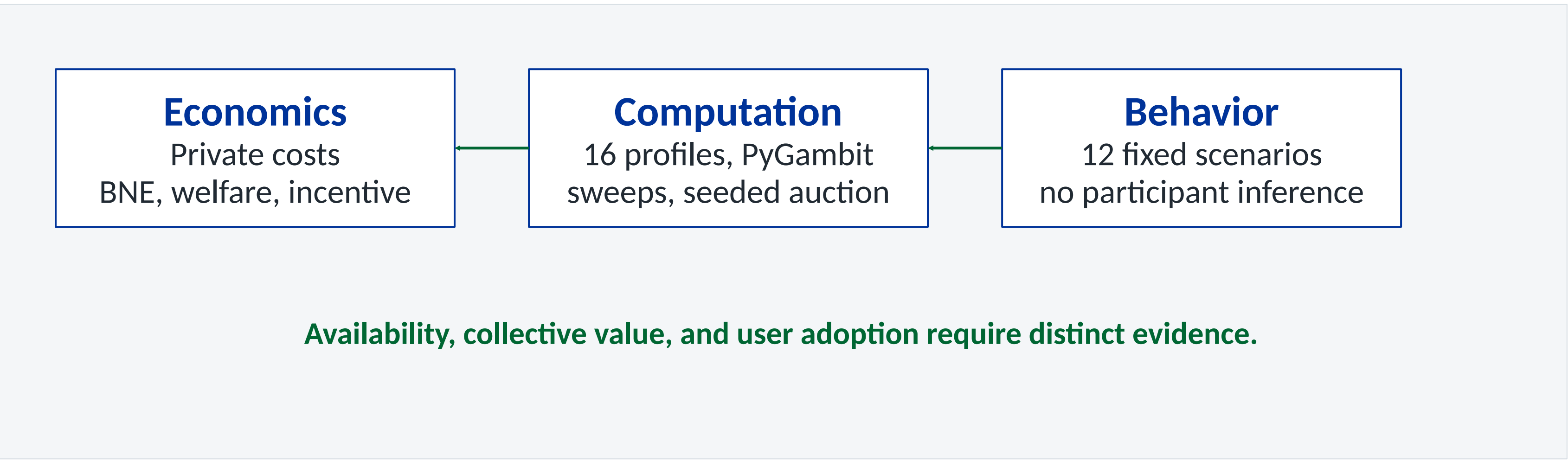


Figure 1. One evidence chain links platform incentives, reproducible verification, and a bounded future user test.

Evidence	Verified output	Status
Model	Pure BNE: (LL,LL) and (PL,PL). Low-type mix: $6/7$.	Analytical and computed
Welfare	At $m=1$: $WLL=0$, $WPL=2.17$, $WPP=4$.	PP is a comparator
Incentive	At $p=.4$, $\tau=.4$ sustains PL. LL remains weak through $\tau=1.2$.	Threshold sweep
Auction	Allocation is .75193 at $r=.50$ and .96004 at $r=.20$.	$N=100,000$, seed 20603

Table 1. Verified outputs at GitHub snapshot f9523b0.
Interpretation: Portability can improve the collective comparator and allocation access while equilibrium multiplicity remains.
Verification: PyGambit agreement, 85/85 tests, and seed 20603.
Evidence status: Platform results are analytical or simulated. The user test has no participant data.

03 Impact & dialogue

Practical impact: a real-world case

Case: A student or writer moving years of assistant context between platforms.

Decision: Whether institutions and platforms should support secure export and import.

Potential benefit: Lower switching friction and greater contestability.

Risks: Privacy leakage, incomplete transfer, and inaccessible tools.

Evidence before adoption: Transfer fidelity, security, access, and switching outcomes.

Interdisciplinary evidence

Behavioral science: Hugging Face
Twelve fixed scenarios compare initial and final Stay/Switch choices around $g>r$. Six predict each choice. The Space stores page-session aggregates only. No participant conclusion.

Computer science: GitHub
Direct enumeration, a PyGambit cross-check, integer-built sweeps, a seeded auction simulation, and 85/85 tests make every result inspectable.

Integration
The public artifact tests whether modeled portability translates into user mobility without changing the platform game.

Limitations & next test

Main limit: Normalized costs and reserves are assumptions, not market estimates. The public Space has no participant dataset.

Next test: Compare switching across portability scenarios while measuring privacy and personalization concerns.

Symposium question: Which omitted concern is most likely to overturn the benchmark?

SEP 28 Symposium, peer evaluation & case impact
SEP 30 Final update

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Selected references

[1] Farrell, J., & Klemperer, P. (2007). Coordination and lock-in. Handbook of Industrial Organization. doi:10.1016/S1573-448X(06)03031-7
[2] Harsanyi, J. C. (1968). Games with incomplete information, II. Management Science, 14(5), 320-334. doi:10.1287/mnsc.14.5.320
[3] Jeon, D.-S., Menicucci, D., & Nasr, N. (2023). Compatibility choices, switching costs, and data portability. AEJ Microeconomics, 15(1), 30-73. doi:10.1257/mic.20200309
[4] Vickrey, W. (1961). Counterspeculation, auctions, and competitive sealed tenders. Journal of Finance, 16(1), 8-37. doi:10.1111/j.1540-6261.1961.tb02789.x

OPEN MATERIALS

Hugging Face: <https://huggingface.co/spaces/mickeystk/ps2-stay-or-switch-memory-portability>
GitHub: <https://github.com/mic1222/gt-tools-demos/tree/ps2-memory-portability>

